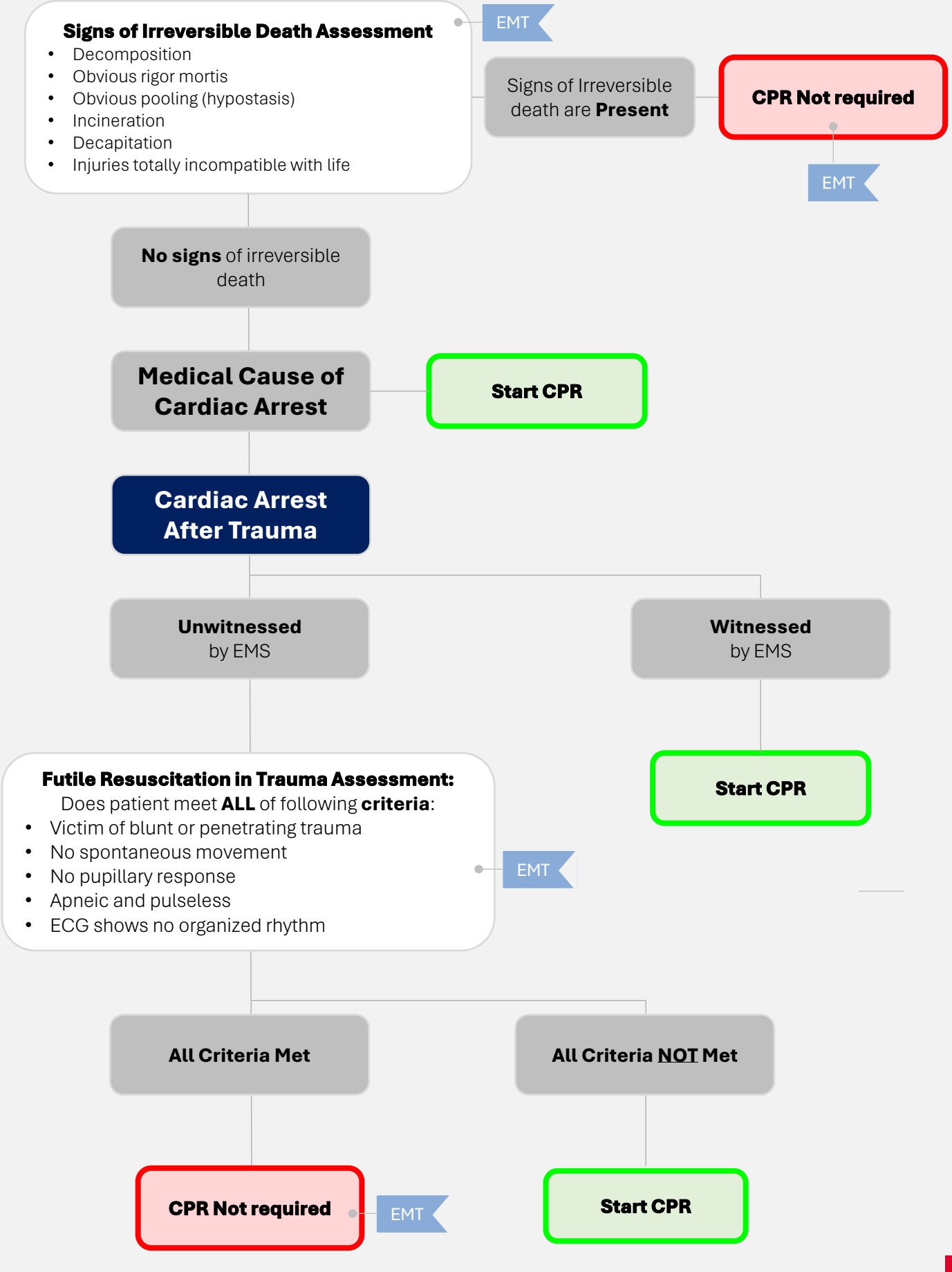
Starting CPR



Transferring CPR to Hospital

CPR Started

Mandatory Cardiac Arrest Transfer Criteria:
Once started, it is mandatory that the below patients are transferred to hospital with on-going CPR:

- Age <18 years Old
- Drowning Cases
- Hypothermia
- Pregnancy
- A clearly identified reversible cause of cardiac arrest that cannot be corrected in the prehospital setting.

Mandatory Transfer **Criteria Met**

Initial Resuscitation Efforts Should Be Performed Prior to Transportation
(e.g., airway management, BVM ventilation, IV access and medication administration).

Transportation to hospital should be prioritized in patients where a fast arrival at hospital will provide clear benefit to the patient (e.g. cardiac arrest secondary to blood loss).

Mandatory Transfer **Criteria NOT Met**

Continue CPR
For at least 20 minutes

Stopping CPR

20 minutes of CPR Completed

Cardiac Arrest After Trauma

Under normal circumstances**, once CPR has been started in the traumatic cardiac arrest patient**:

- **The patient should be transferred to hospital**
- On scene time should be minimized

In uncommon circumstances (such as difficult extrications) Paramedics (and higher clinical level) may stop resuscitation efforts in the trauma patient, according the criteria listed for Medical Cardiac Arrest below.

P

Medical Cardiac Arrest

Paramedics (and higher clinical level) may stop resuscitation efforts in the non-trauma patient if:

- The patient is not included in Mandatory Cardiac Arrest Transfer Criteria:
  - Age <18 years Old
  - Drowning Cases
  - Hypothermia
  - Pregnancy
  - A clearly identified reversible cause of cardiac arrest that cannot be corrected in the prehospital setting.
- No pulse, AND/
- No respirations, AND/
- Appropriate ALS resuscitation interventions for the cardiac rhythm has been administered, AND/
- No ROSC has been achieved after 20 minutes of resuscitation, AND/
- The cardiac rhythm is asystole, AND/
- EtCO2 is less than 20mmHg
- No heart sounds, AND/
- No palpable pulse (carotid or femoral), AND/
- No evidence of neurological function (non-reactive pupils, no response to pain, no spontaneous movement), AND/
- All clinicians involved in the patient's care agree that stopping the resuscitation is appropriate.

P

Patient Care Goals

1. Promptly initiate high-quality CPR (resuscitation) for patients in cardiac arrest.
2. Make informed decisions about stopping or transferring CPR based on clinical criteria and futility to avoid unnecessary prolongation of resuscitation.
3. Maintain effective CPR during transfer.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with cardiac arrest, including:
  - Unresponsiveness with no signs of life (e.g., no pulse, no breathing, or agonal gasps).
  - Suspected pulseless electrical activity (PEA), ventricular fibrillation (VF), ventricular tachycardia (VT), or asystole.
- Patients requiring decisions about initiating, continuing, or terminating CPR.

Exclusion Criteria

- None Noted.

Key Patient Assessment Points

1. **Cardiac Arrest Confirmation**
   - Verify unresponsiveness by assessing for response to verbal or painful stimuli (AVPU: U).
   - Check for absence of pulse (carotid or femoral) for 5-10 seconds.
   - Confirm absence of normal breathing or presence of agonal gasps.
2. **Bystander CPR**
   - Note the presence (or absence) of bystander CPR; including bystander CPR quality.
3. **Cardiac Rhythm**
   - Apply cardiac monitor to identify rhythm (VF/VT/PEA/Asystole).
4. **Time Patient Last Seen Alive**
   - Note time of arrest onset or when patient was last seen alive.
5. **Signs of Irreversible Death**
   - Assess for signs that indicate incompatibility with life:
     - Decomposition
     - Obvious rigor mortis
     - Obvious pooling (hypostasis)
     - Incineration
     - Decapitation
     - Injuries totally incompatible with life
6. **Futile Resuscitation in Trauma Assessment**
   - In the unwitnessed (by EMS) traumatic cardiac arrest patient, assess for signs which indicate that resuscitation is not indicated. Including:
     - Victim of blunt or penetrating trauma
     - No spontaneous movement
     - No pupillary response
     - Apneic and pulseless
     - ECG shows no organized rhythm

Patient Management & Interventions

1. **Starting CPR**
   - CPR does not need to be started on patients who meet the criteria listed below:
     - **Signs of Irreversible Death Assessment**
       - CPR/Resuscitation does not need to be started if any <u>one</u> of the below signs are present:
         - Decomposition
         - Obvious rigor mortis

            - Obvious pooling (hypostasis)
            - Incineration
            - Decapitation
            - Injuries totally incompatible with life
        - If it is not clear if a sign of irreversible death is present, CPR should be started.
    - **Futile Resuscitation in Trauma Assessment**
        - In the unwitnessed (by EMS) traumatic cardiac arrest patient, CPR does not need to be started for patients that meet the below criteria:
            - Victim of blunt or penetrating trauma
            - No spontaneous movement
            - No pupillary response
            - Apneic and pulseless
            - ECG shows no organized rhythm
        - All signs must be present to not start resuscitation (CPR)
- CPR should be started on all other patients.
- Once started, CPR should generally be continued for at least 20 minutes.

2. **Transferring CPR**
    - Once started, it is mandatory that the below patients are transferred to hospital with on-going CPR:
        - **Mandatory Cardiac Arrest Transfer Criteria:**
            - Age <18 years Old
            - Drowning Cases
            - Hypothermia
            - Pregnancy
            - A clearly identified reversible cause of cardiac arrest that cannot be corrected in the prehospital setting.
    - Initial resuscitation efforts should be performed prior to transportation (e.g., airway management, BVM ventilation, IV access and medication administration).
    - Transportation to hospital should be prioritized in patients where a fast arrival at hospital will provide clear benefit to the patient (e.g. cardiac arrest secondary to blood loss).
    - If the patients' family requests the patient be transferred to hospital, it is appropriate to transfer the patient to hospital.
3. **Stopping CPR**
    - Once CPR has been started, CPR/resuscitation efforts may only be stopped by a Paramedic, Advanced Paramedic or Emergency Physician.
        - EMT's cannot stop resuscitation efforts. EMT's must either transfer the patient to hospital or wait for ALS back-up.
        - If ALS back-up is significantly far away and additional staff are required to ensure effective CPR during extrication and transfer, another BLS ambulance should be requested if required.
4. **Stopping CPR in the non-trauma patient**
    - Paramedics (and higher clinical level) may stop resuscitation efforts in the **non-trauma patient** if:
        - The patient is not included in **Mandatory Cardiac Arrest Transfer Criteria**:
            - Age <18 years Old
            - Drowning Cases
            - Hypothermia
            - Pregnancy
            - A clearly identified reversible cause of cardiac arrest that cannot be corrected in the prehospital setting.
        - No pulse, AND/
        - No respirations, AND/
        - Appropriate ALS resuscitation interventions for the cardiac rhythm has been administered, AND/
        - No ROSC has been achieved after 20 minutes of resuscitation, AND/

- The cardiac rhythm is asystole, AND/
- EtCO2 is less than 20mmHg
- No heart sounds, AND/
- No palpable pulse (carotid or femoral), AND/
- No evidence of neurological function (non-reactive pupils, no response to pain, no spontaneous movement), AND/
- All clinicians involved in the patient's care agree that stopping the resuscitation is appropriate.

- If the resuscitation is not terminated, the patient should be transferred to hospital

5. **Stopping CPR in the trauma patient**
   - Under normal circumstances, in the traumatic cardiac arrest patient, if CPR is started, transfer to hospital should be initiated and time on-scene minimized.
   - In uncommon circumstances (such as difficult extrications) Paramedics (and higher clinical level) may stop resuscitation efforts in the **trauma patient** if:
     - The patient is not included in **Mandatory Cardiac Arrest Transfer Criteria**:
       - Age <18 years Old
       - Drowning Cases
       - Hypothermia
       - Pregnancy
       - A clearly identified reversible cause of cardiac arrest that cannot be corrected in the prehospital setting.
     - No pulse, AND/
     - No respirations, AND/
     - Appropriate ALS resuscitation interventions for the cardiac rhythm has been administered, AND/
     - No ROSC has been achieved after 20 minutes of resuscitation, AND/
     - The cardiac rhythm is asystole, AND/
     - EtCO2 is less than 20mmHg
     - No heart sounds, AND/
     - No palpable pulse (carotid or femoral), AND/
     - No evidence of neurological function (non-reactive pupils, no response to pain, no spontaneous movement), AND/
     - All clinicians involved in the patient's care agree that stopping the resuscitation is appropriate.
   - If the resuscitation is not terminated, the patient should be transferred to hospital

Patient Safety Considerations

- If there is any doubt surrounding starting CPR, CPR should be started.
- Transportation to hospital should be prioritized in patients where a fast arrival at hospital will provide clear benefit to the patient (e.g. cardiac arrest secondary to blood loss).
- ALS Back Up should be routinely requested for all patients undergoing CPR.
- If a patient is to be transferred to hospital under CPR, effective resuscitation must be maintained during extrication and transfer (chest compressions, airway management, ventilation, medication administration).

Clinical Insights & Additional Guidance

- Providing Treatment Prior to Transfer
  - In most cases, appropriate initial resuscitation should be performed on-scene prior to starting transfer. This includes:
    - Airway management
    - Bag Valve Mask Ventilation
    - Chest Compressions
    - Defibrillation (if required)
    - IV/IO Access and appropriate medication administration
    - Reversal of any identified causes of cardiac arrest

- If it is difficult to perform some procedures (e.g. IV access) prior to transfer, a clinical decision must be made by the staff on scene. This decision must balance the time taken to perform the procedure vs. the time taken to effectively perform the procedure.

Documentation

- Identification: Full patient name, age, and gender.
- Mechanism / Medical Complaint: Cardiac arrest type, cause (if known), and whether CPR was witnessed or unwitnessed.
- Injuries / Information: Any trauma or clinical signs incompatible with life; bystander CPR provided; last seen alive time.
- Signs: Initial rhythm, $EtCO_2$ values, response to CPR or ALS interventions, ROSC signs (if any).
- Treatment:
    - Time CPR started and stopped.
    - Defibrillation details if applicable.
    - Medications administered (type, dose, route, time).
    - Ventilation method and airway management.
    - Transfer decision rationale (e.g., futility or reversible cause).
- Allergies: Document if known.
- Medications: Any available pre-arrest medications.
- Background: Medical history, trauma, or hypothermia indicators.
- Other: Scene conditions, staff involved, equipment used (e.g., mechanical CPR).
    - Consensus notes for stopping CPR (if applicable).
    - Ensure accurate time-stamping of key actions (CPR start/stop, medication, rhythm changes).
    - Clearly document rationale for transferring or stopping CPR.
    - If CPR is stopped, include signs of irreversible death or all futility criteria met.

Key Performance Indicators

1. Ensure all relevant information surrounding the decision to start/stop or transfer CPR is captured in the patient care report.

References

1. American Heart Association 2020, *2020 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care*, American Heart Association, Dallas.
2. Ambulance Victoria 2023, *Clinical Practice Guidelines for Ambulance and MICA Paramedics*, Ambulance Victoria, Melbourne.
3. Anderson, N.E., Parry, D., Beever, D. & Smith, J. 2021, 'Termination of resuscitation in out-of-hospital cardiac arrest: A review of current guidelines and evidence', *Emergency Medicine Journal*, vol. 38, no. 6, pp. 423–429.
4. British Columbia Emergency Health Services 2020, *British Columbia Emergency Health Services Patient Care Guidelines*, BC Emergency Health Services, Vancouver.
5. Drennan, I.R., Case, E., Verbeek, P.R., Reynolds, J.C., Goldberger, Z.D., Jasti, J., Charleston, M., Herren, H., Idris, A.H., Leslie, P.R., Austin, M.A., Xiong, Y., Schmicker, R.H., Stub, D., Morrison, L.J. & Resuscitation Outcomes Consortium Investigators 2017, 'A comparison of the universal TOR guideline to the absence of out-of-hospital TOR guideline in two ground EMS systems', *Resuscitation*, vol. 111, pp. 45–51.
6. Elling, B., Smith, M. & Pollak, A.N. 2018, *Nancy Caroline's Emergency Care in the Streets*, 8th edn, Jones & Bartlett Learning, Burlington, MA.
7. Joint Royal Colleges Ambulance Liaison Committee 2019, *JRCALC Clinical Guidelines 2019*, Class Professional Publishing, Bridgewater.
8. Kramer, E.B., Botha, M., Dunning, J., Wallis, L.A., Deakin, C.D. & Perkins, G.D. 2018, 'Part 6: Alternative techniques and ancillary devices for cardiopulmonary resuscitation: 2015 International Consensus on Cardiopulmonary Resuscitation and Emergency Cardiovascular Care Science with Treatment Recommendations', *Resuscitation*, vol. 95, pp. 150–171.

9. Lockey, D.J., Lyon, R.M., Davies, G.E., Coats, T.J., Russell, M.Q., Dark, P., Caputo, M.L., Emsley, J., Benger, J. & Nolan, J.P. 2014, 'Termination of resuscitation in traumatic cardiac arrest: A novel decision-making algorithm', *Resuscitation*, vol. 85, no. 10, pp. 1378–1383.
10. Morrison, L.J., Visentin, L.M., Kiss, A., Theriault, R., Ewen, M., Vermeulen, M., Sherbino, J., Verbeek, P.R. & TOR Investigators 2006, 'Validation of a rule for termination of resuscitation in out-of-hospital cardiac arrest', *New England Journal of Medicine*, vol. 355, no. 5, pp. 478–487.
11. National Association of State EMS Officials 2022, *National Model EMS Clinical Guidelines*, Version 3, NASEMSO, Falls Church, VA.
12. New South Wales Ambulance 2022, *Clinical Practice Guidelines: Medical Management of Patients*, NSW Ambulance, Sydney.
13. New Zealand Ambulance Service 2021, *Clinical Procedures and Guidelines: Comprehensive Edition 2021–2023*, St John New Zealand, Auckland.
14. Ontario Ministry of Health and Long-Term Care 2020, *Advanced Life Support Patient Care Standards*, Version 4.9, Emergency Health Services Branch, Toronto.
15. Paramedic Association of Canada 2021, *National Occupational Competency Profile for Paramedics*, Paramedic Association of Canada, Ottawa.
16. Perkins, G.D., Ji, C., Deakin, C.D., Quinn, T., Nolan, J.P., Scomparin, C., Regan, S., Long, J., Slowther, A., Pocock, H., Black, J.J.M., Gates, S., Lall, R. & PARAMEDIC2 Collaborators 2021, 'A randomized trial of epinephrine in out-of-hospital cardiac arrest', *New England Journal of Medicine*, vol. 384, no. 11, pp. 877–886.
17. Resuscitation Council UK 2021, *Resuscitation Guidelines 2021*, Resuscitation Council UK, London.
18. Sandroni, C., Cariou, A., Cavallaro, F., Cronberg, T., Friberg, H., Hoedemaekers, C., Horn, J., Nolan, J.P., Rossetti, A.O. & Soar, J. 2014, 'Prognostication in comatose survivors of cardiac arrest: An advisory statement from the European Resuscitation Council and the European Society of Intensive Care Medicine', *Resuscitation*, vol. 85, no. 12, pp. 1779–1789.
19. Soar, J., Nolan, J.P., Böttiger, B.W., Perkins, G.D., Lott, C., Carli, P., Pellis, T., Sandroni, C., Skrifvars, M.B., Smith, G.B., Sunde, K. & Deakin, C.D. 2021, 'European Resuscitation Council Guidelines 2021: Section 3. Adult advanced life support', *Resuscitation*, vol. 161, pp. 115–151.
20. St John Ambulance Western Australia 2023, *Clinical Practice Guidelines: Cardiac Arrest Management*, St John WA, Perth.
21. Stub, D., Bernard, S., Pellegrino, V., Smith, K., Walker, T., Sheldrake, J., Hockings, L., Shaw, J., Duffy, S.J., Burrell, A., Cameron, P., Smit, D.V. & Kaye, D.M. 2015, 'Refractory cardiac arrest treated with mechanical CPR, hypothermia, ECMO and early reperfusion (the CHEER trial)', *Resuscitation*, vol. 86, pp. 88–94.
22. Tomasello, S., Sandroni, C. & Peris, A. 2020, 'Termination of resuscitation in out-of-hospital cardiac arrest: A systematic review', *Critical Care*, vol. 24, no. 1, p. 251.
23. Tokyo Fire Department 2021, *Tokyo Emergency Medical Service System Guidelines*, Tokyo Fire Department, Tokyo.
24. Truhlar, A., Deakin, C.D., Soar, J., Khalifa, G.E.A., Alfonzo, A., Bierens, J.J.L.M., Brattebø, G., Brugger, H., Dunning, J., Hunyadi-Antičević, S., Koster, R.W., Lockey, D.J., Lott, C., Paal, P., Perkins, G.D., Sandroni, C. & Nolan, J.P. 2015, 'European Resuscitation Council Guidelines for Resuscitation 2015: Section 4. Cardiac arrest in special circumstances', *Resuscitation*, vol. 95, pp. 148–201.
25. Vanden Hoek, T.L., Morrison, L.J., Shuster, M., Donnino, M., Sinz, E., Lavonas, E.J., Jeejeebhoy, F.M. & Gabrielli, A. 2010, 'Part 12: Cardiac arrest in special situations: 2010 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care', *Circulation*, vol. 122, no. 18 Suppl 3, pp. S829–S861.
26. Weingart, S.D. & Levitan, R.M. 2011, 'Prehospital management of airway emergencies', *Emergency Medicine Clinics of North America*, vol. 29, no. 4, pp. 709–730.
27. Wong, M.K., Morrison, L.J., Qiu, F., Austin, P.C., Cheskes, S., Dorian, P., Welsford, M., Brooks, S.C., Stiell, I.G., Brown, S.P., Christenson, J., Resuscitation Outcomes Consortium Investigators 2014, 'Trends in in-hospital cardiopulmonary resuscitation and survival in adults between 1993 and 2012: A study of the American Heart Association's Get With The Guidelines-Resuscitation registry', *JAMA Cardiology*, vol. 3, no. 3, pp. 214–222.

28. Zive, D., Daya, M., Schmicker, R., Powell, J., Jui, J., Kusch, J., Buick, J.E., Morrison, L.J., Resuscitation Outcomes Consortium Investigators 2017, 'Implementation of the TOR rule in the out-of-hospital setting: Impact on survival in a rural EMS system', *Prehospital Emergency Care*, vol. 21, no. 4, pp. 478–483.